

```
%% =====
```

```
%% PROBLEM P7-6A: Reaction A → B + C
```

```
%% =====
```

```
fprintf('=== PROBLEM P7-6A ===\n\n');
```

```
% Data
```

```
t_6a = [0, 5, 9, 15, 22, 30, 40, 60]; % time in minutes
```

```
CA = [2, 1.6, 1.35, 1.1, 0.87, 0.70, 0.53, 0.35]; % concentration in mol/dm3
```

```
%% (a) Determine reaction order, alpha, and specific reaction rate k
```

```
fprintf('(a) Determining reaction order and rate constant:\n\n');
```

```
% Method 1: Nonlinear Least Squares Regression
```

```
% Test orders: 0, 1, 2
```

```
% Zero-order: CA = CA0 - k*t
```

```
% First-order: ln(CA) = ln(CA0) - k*t
```

```
% Second-order: 1/CA = 1/CA0 + k*t
```

```
% Zero-order model
```

```
p0 = polyfit(t_6a, CA, 1);
```

```
k0 = -p0(1);
```

```
CA_pred0 = polyval(p0, t_6a);
```

```
SSE0 = sum((CA - CA_pred0).^2);
```

```
R2_0 = 1 - SSE0/sum((CA - mean(CA)).^2);
```

```
% First-order model
```

```
p1 = polyfit(t_6a, log(CA), 1);
```

```
k1 = -p1(1);
```

```
CA_pred1 = exp(polyval(p1, t_6a));
```

```
SSE1 = sum((CA - CA_pred1).^2);
```

```
R2_1 = 1 - SSE1/sum((CA - mean(CA)).^2);
```

```
% Second-order model
```

```
p2 = polyfit(t_6a, 1./CA, 1);
```

Command Window

```
=== PROBLEM P7-6A ===
```

```
(a) Determining reaction order and rate constant:
```




```

% Second-order model
p2 = polyfit(t_6a, 1./CA, 1);
k2 = p2(1);
CA_pred2 = 1./polyval(p2, t_6a);
SSE2 = sum((CA - CA_pred2).^2);
R2_2 = 1 - SSE2/sum((CA - mean(CA)).^2);

% Display results
fprintf('Method 1: Linear Regression\n');
fprintf('Zero-order: k = %.4f mol/dm3.min), R2 = %.6f, SSE = %.6f\n', k0, R2_0, SSE0);
fprintf('First-order: k = %.4f min-1, R2 = %.6f, SSE = %.6f\n', k1, R2_1, SSE1);
fprintf('Second-order: k = %.4f dm3/(mol.min), R2 = %.6f, SSE = %.6f\n\n', k2, R2_2, SSE2);

```

```

% Method 2: Differential Method
% Calculate -dCA/dt using finite differences
dCA_dt = -diff(CA)./diff(t_6a);
CA_avg = (CA(1:end-1) + CA(2:end))/2;

% Plot log(-dCA/dt) vs log(CA) - slope gives order
log_rate = log(dCA_dt);
log_CA = log(CA_avg);
p_diff = polyfit(log_CA, log_rate, 1);
alpha_diff = p_diff(1);
k_diff = exp(p_diff(2));

fprintf('Method 2: Differential Method\n');
fprintf('Reaction order α = %.2f\n', alpha_diff);
fprintf('Rate constant k = %.4f (units depend on order)\n\n', k_diff);

% Determine best order
[~, best_order_idx] = max([R2_0, R2_1, R2_2]);
orders = [0, 1, 2];
best_order = orders(best_order_idx);
k_values = [k0, k1, k2];
best_k = k_values(best_order_idx);

fprintf('CONCLUSION: Best fit is order α = %d with k = %.4f\n\n', best_order, best_k);

```

Command Window

=== PROBLEM P7-6A ===

(a) Determining reaction order

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```
best_k = k_values(best_order_idx);
```

```
fprintf('CONCLUSION: Best fit is order a = %d with k = %.4f\n\n', best_order, best_k);
```

```
% Plotting
```

```
figure('Position', [100, 100, 1400, 900]);
```

```
subplot(2,3,1)
```

```
plot(t_6a, CA, 'ko', 'MarkerSize', 8, 'LineWidth', 1.5); hold on;
```

```
plot(t_6a, CA_pred0, 'b-', 'LineWidth', 2);
```

```
xlabel('Time (min)'); ylabel('C_A (mol/dm3)');
```

```
title(sprintf('Zero-Order: R2 = %.4f', R2_0));
```

```
legend('Data', 'Fit', 'Location', 'best'); grid on;
```

```
subplot(2,3,2)
```

```
plot(t_6a, CA, 'ko', 'MarkerSize', 8, 'LineWidth', 1.5); hold on;
```

```
plot(t_6a, CA_pred1, 'r-', 'LineWidth', 2);
```

```
xlabel('Time (min)'); ylabel('C_A (mol/dm3)');
```

```
title(sprintf('First-Order: R2 = %.4f', R2_1));
```

```
legend('Data', 'Fit', 'Location', 'best'); grid on;
```

```
subplot(2,3,3)
```

```
plot(t_6a, CA, 'ko', 'MarkerSize', 8, 'LineWidth', 1.5); hold on;
```

```
plot(t_6a, CA_pred2, 'g-', 'LineWidth', 2);
```

```
xlabel('Time (min)'); ylabel('C_A (mol/dm3)');
```

```
title(sprintf('Second-Order: R2 = %.4f', R2_2));
```

```
legend('Data', 'Fit', 'Location', 'best'); grid on;
```

```
subplot(2,3,4)
```

```
plot(t_6a, CA, 'b-', 'LineWidth', 2);
```

```
xlabel('Time (min)'); ylabel('C_A (mol/dm3)');
```

```
title('Zero-Order: C_A vs t'); grid on;
```

```
subplot(2,3,5)
```

```
plot(t_6a, log(CA), 'r-', 'LineWidth', 2);
```

```
xlabel('Time (min)'); ylabel('ln(C_A)');
```

```
title('First-Order: ln(C_A) vs t'); grid on;
```



```
sgtitle('P7-6A: Reaction Order Analysis');
```

```
%% (b) Where to place additional data points?
```

```
fprintf('(b) Optimal placement of additional data points:\n');
```

```
fprintf('    - Early times (0-10 min): Better resolution of initial rate\n');
```

```
fprintf('    - Around t = 15-25 min: Maximum curvature region\n');
```

```
fprintf('    - Late times (> 60 min): Verify approach to completion\n');
```

```
fprintf('    Recommended: t = 2, 7, 18, 25, 50, 80 min\n\n');
```

```
%% (c) If repeating experiment at different temperature
```

```
fprintf('(c) Repeating experiment at different temperaturee:\n');
```

```
fprintf('    - Run at HIGHER temperature (increases k, faster reaction)\n');
```

```
fprintf('    - Use same time intervals initially, then adjust based on reaction speed\n');
```

```
fprintf('    - If much faster: take more early points (0-30 min range)\n');
```

```
fprintf('    - If much slower: extend time range beyond 60 min\n');
```

```
fprintf('    - At least 3 temperatures needed for Arrhenius analysis\n\n');
```

```
%% (d) Effect of dilution error at t = 60 min
```

```
fprintf('(d) Dilution error at t = 60 min:\n');
```

```
fprintf('    If C_A(60) = 0.35 has dilution error (likely too high):\n');
```

```
fprintf('    - Polynomial/graphical methods: Last point pulls curve up\n');
```

```
fprintf('    - Regression methods: Weighted by all points, less sensitive\n');
```

```
fprintf('    - Would increase apparent R2 for zero-order (wrong)\n');
```

```
fprintf('    - Would decrease R2 for higher orders (correct indication)\n');
```

```
fprintf('    - Could lead to wrong order determination if borderline\n\n');
```

```
% Simulate effect of dilution error
```

```
CA_error = CA;
```

```
CA_error(end) = 0.45; % Assume 30% dilution error
```

```
p1_error = polyfit(t_6a, log(CA_error), 1);
```

```
CA_pred1_error = exp(polyval(p1_error, t_6a));
```

```
SSE1_error = sum((CA_error - CA_pred1_error).^2);
```

```
R2_1_error = 1 - SSE1_error/sum((CA_error - mean(CA_error)).^2);
```


=== PROBLEM P7-6A ===

(a) Determining reaction order and rate constant:

Method 1: Linear Regression

Zero-order: $k = 0.0260 \text{ mol/dm}^3 \cdot \text{min}$, $R^2 = 0.862081$, $\text{SSE} = 0.305015$

First-order: $k = 0.0289 \text{ min}^{-1}$, $R^2 = 0.970746$, $\text{SSE} = 0.064697$

Second-order: $k = 0.0390 \text{ dm}^3/(\text{mol} \cdot \text{min})$, $R^2 = 0.802348$, $\text{SSE} = 0.437118$

Method 2: Differential Method

Reaction order $\alpha = 1.53$

Rate constant $k = 0.0327$ (units depend on order)

CONCLUSION: Best fit is order $\alpha = 1$ with $k = 0.0289$

(b) Optimal placement of additional data points:

- Early times (0-10 min): Better resolution of initial rate
- Around $t = 15-25 \text{ min}$: Maximum curvature region
- Late times ($> 60 \text{ min}$): Verify approach to completion

Recommended: $t = 2, 7, 18, 25, 50, 80 \text{ min}$

(c) Repeating experiment at different temperatures:

- Run at HIGHER temperature (increases k , faster reaction)
- Use same time intervals initially, then adjust based on reaction speed
- If much faster: take more early points (0-30 min range)
- If much slower: extend time range beyond 60 min
- At least 3 temperatures needed for Arrhenius analysis

(d) Dilution error at $t = 60 \text{ min}$:

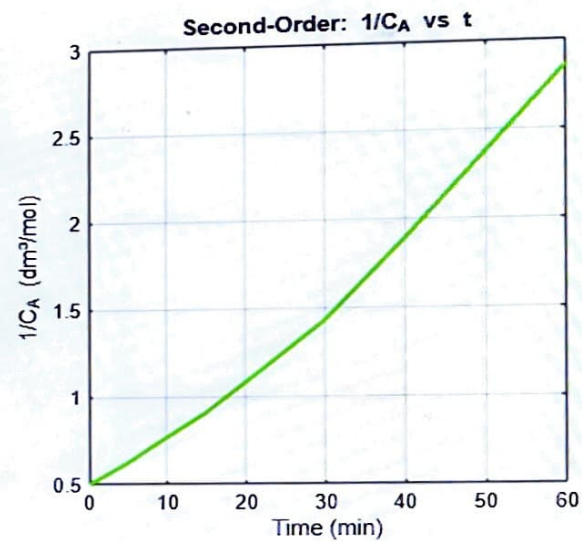
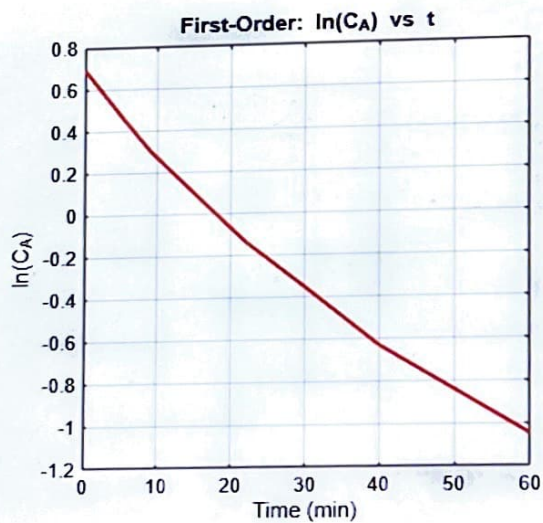
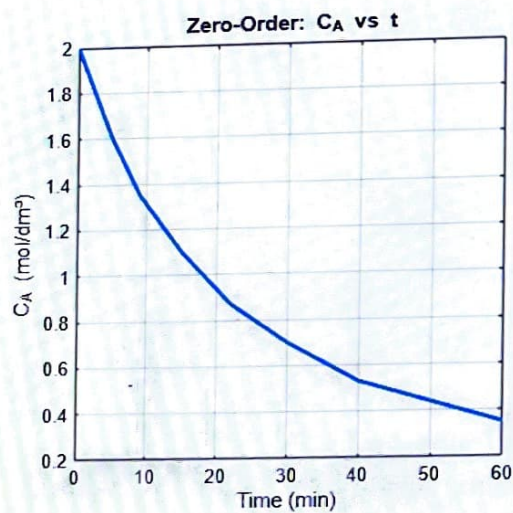
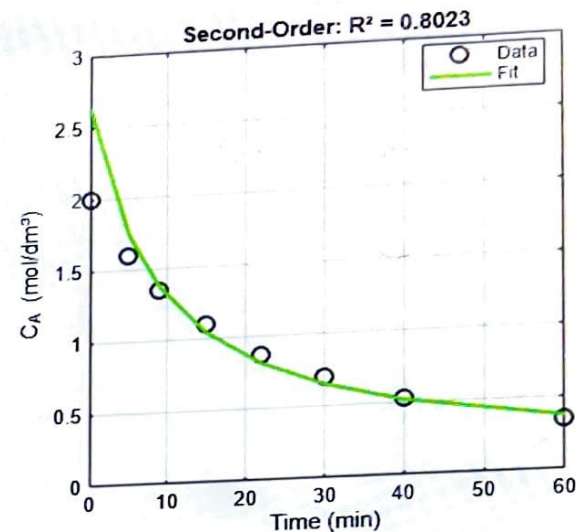
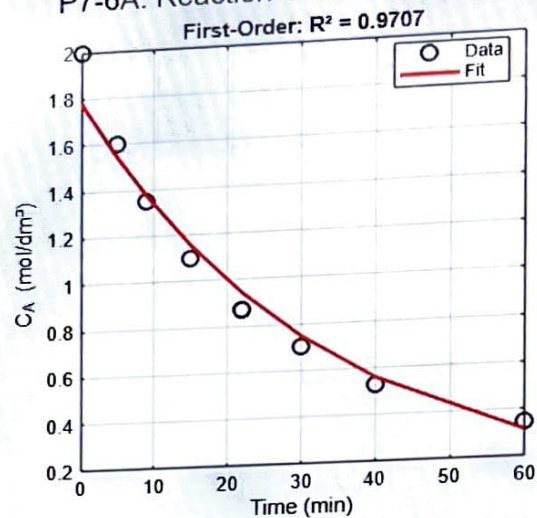
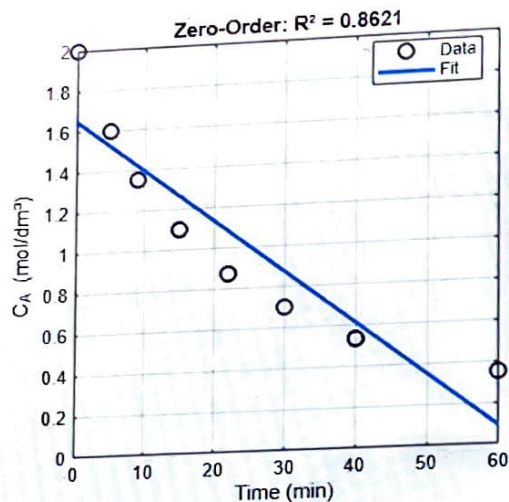
If $C_A(60) = 0.35$ has dilution error (likely too high):

- Polynomial/graphical methods: Last point pulls curve up
- Regression methods: Weighted by all points, less sensitive
- Would increase apparent R^2 for zero-order (wrong)
- Would decrease R^2 for higher orders (correct indication)
- Could lead to wrong order determination if borderline

Example: If actual $C_A(60) = 0.45$ due to dilution:

First-order R^2 drops from 0.9707 to 0.9358

P7-6A: Reaction Order Analysis



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150

151 %% =====

152 %% **PROBLEM P7-7A: Decomposition of dimethyl ether**

153 %% =====

154

155 fprintf('\n=== PROBLEM P7-7A ===\n\n');

156 fprintf('Reaction: $(\text{CH}_3)_2\text{O} \rightarrow \text{CH}_4 + \text{H}_2 + \text{CO}$ \n');

157 fprintf('Constant-volume batch reactor, 504°C\n\n');

158

159 % Data

160 t_7a = [390, 777, 1195, 3155]; % time in seconds

161 P_total = [408, 488, 562, 799]; % total pressure in mmHg

162 P_inf = 931; % final pressure at t = infinity

163

164 %% (a) Estimate initial pressure at t = 0

165 fprintf('(a) Estimating initial pressure at t = 0:\n\n');

166

167 % Stoichiometry: 1 mole DME $\rightarrow$ 3 moles products (net: +2 moles)

168 % P_total = P_DME + P_products

169 % At t=0: P_total = P_DME0

170 % At t= ∞ : P_total = P_products = 3*P_DME0

171 % Therefore: P_inf = 3*P_DME0, so P_DME0 = P_inf/3

172

173 P_DME0_estimated = P_inf / 3;

174 fprintf(' Stoichiometry: 1 DME $\rightarrow$ 1 CH₄ + 1 H₂ + 1 CO (net $\Delta n = +2$)\n');175 fprintf(' P(t= ∞) = 3 $\times$ P_DME(t=0)\n');176 fprintf(' P_DME(t=0) = P(t= ∞)/3 = %.1f mmHg\n\n', P_DME0_estimated);

177

178 %% (b) Determine reaction order and rate constant

179 fprintf('(b) Determining reaction order and rate constant:\n\n');

180

181 % Calculate partial pressure of DME at each time

182 % Material balance: P_DME = P_DME0 - (P_total - P_DME0)/2

183 % Or: P_DME = (3*P_DME0 - P_total)/2 = (P_inf - P_total)/2

184

185 P_DME = (P_inf - P_total) / 2;

186 P_DME_all = [P_DME0_estimated, P_DME];

187 t_all = [0, t_7a];

188

189 fprintf('Time (s)\tP_total (mmHg)\tP_DME (mmHg)\n');

190 fprintf('0\t\t%.1f\t\t%.1f\n', P_DME0_estimated, P_DME0_estimated);

191 for i = 1:length(t_7a)




```

188 fprintf('Time (s)\tP_total (mmHg)\tP_DME (mmHg)\n');
189 fprintf('0\t%.1f\t%.1f\n', P_DME0_estimated, P_DME0_estimated);
190 for i = 1:length(t_7a)
191     fprintf('%.0f\t%.0f\t%.2f\n', t_7a(i), P_total(i), P_DME(i));
192 end
193 fprintf('\n');
194
195 % Test reaction orders
196 % First-order:  $\ln(P_{DME}) = \ln(P_{DME0}) - k \cdot t$ 
197 p_1st = polyfit(t_all, log(P_DME_all), 1);
198 k_1st = -p_1st(1);
199 P_DME_pred_1st = exp(polyval(p_1st, t_all));
200 SSE_1st = sum((P_DME_all - P_DME_pred_1st).^2);
201 R2_1st = 1 - SSE_1st/sum((P_DME_all - mean(P_DME_all)).^2);
202
203 % Second-order:  $1/P_{DME} = 1/P_{DME0} + k \cdot t$ 
204 p_2nd = polyfit(t_all, 1./P_DME_all, 1);
205 k_2nd = p_2nd(1);
206 P_DME_pred_2nd = 1./polyval(p_2nd, t_all);
207 SSE_2nd = sum((P_DME_all - P_DME_pred_2nd).^2);
208 R2_2nd = 1 - SSE_2nd/sum((P_DME_all - mean(P_DME_all)).^2);
209
210 fprintf('First-order: k = %.6f s-1, R2 = %.6f\n', k_1st, R2_1st);
211 fprintf('Second-order: k = %.6f mmHg-1·s-1, R2 = %.6f\n\n', k_2nd, R2_2nd);
212
213 if R2_1st > R2_2nd
214     fprintf('CONCLUSION: Reaction is FIRST-ORDER\n');
215     fprintf('Rate constant k = %.6f s-1 (%.4e s-1)\n', k_1st, k_1st);
216     fprintf('Half-life t1/2 = ln(2)/k = %.1f s\n\n', log(2)/k_1st);
217 else
218     fprintf('CONCLUSION: Reaction is SECOND-ORDER\n');
219     fprintf('Rate constant k = %.6f mmHg-1·s-1\n\n', k_2nd);
220 end
221
222 % Plotting
223 figure('Position', [100, 100, 1400, 500]);
224
225 subplot(1,3,1)
226 plot(t_all, P_DME_all, 'ko', 'MarkerSize', 10, 'LineWidth', 2); hold on;
227 t_fit = linspace(0, max(t_all), 100);
228 plot(t_fit, exp(polyval(p_1st, t_fit)), 'b-', 'LineWidth', 2);

```


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229 plot(t_fit, exp(polyval(p_1st, t_fit)), 'b-', 'LineWidth', 2);
230 xlabel('Time (s)'); ylabel('P_{DME} (mmHg)');
231 title('DME Pressure vs Time'); grid on; legend('Data', 'First-order fit');
232
233 subplot(1,3,2)
234 plot(t_all, log(P_DME_all), 'ro', 'MarkerSize', 10, 'LineWidth', 2); hold on;
235 plot(t_fit, polyval(p_1st, t_fit), 'r-', 'LineWidth', 2);
236 xlabel('Time (s)'); ylabel('ln(P_{DME})');
237 title(sprintf('First-Order: R^2 = %.6f', R2_1st)); grid on;
238
239 subplot(1,3,3)
240 plot(t_all, 1./P_DME_all, 'go', 'MarkerSize', 10, 'LineWidth', 2); hold on;
241 plot(t_fit, polyval(p_2nd, t_fit), 'g-', 'LineWidth', 2);
242 xlabel('Time (s)'); ylabel('1/P_{DME} (mmHg^{-1})');
243 title(sprintf('Second-Order: R^2 = %.6f', R2_2nd)); grid on;
244
245 sgtitle('P7-7A: Dimethyl Ether Decomposition');
246

```

%% (c) Experimental conditions for more data

```

247 fprintf('(c) Suggested experimental conditions for more data:\n');
248 fprintf('    - More points at early times (50-500 s): Better initial rate\n');
249 fprintf('    - Intermediate times (1000-2000 s): Verify linearity\n');
250 fprintf('    - Use higher precision pressure gauge\n');
251 fprintf('    - Verify constant volume and temperature control\n');
252 fprintf('    - Replicate measurements to assess uncertainty\n');
253 fprintf('    - Check for side reactions or catalytic effects\n\n');
254

```

%% (d) Effect of higher temperature

```

256 fprintf('(d) Running at higher temperature:\n');
257 fprintf('    DATA CHANGES:\n');
258 fprintf('    - Reaction rate k increases exponentially (Arrhenius)\n');
259 fprintf('    - Same pressures reached at earlier times\n');
260 fprintf('    - P(t=∞) = 931 mmHg unchanged (equilibrium position)\n');
261 fprintf('    - Would need shorter time intervals\n\n');
262 fprintf('    ANSWER CHANGES:\n');
263 fprintf('    - Reaction order: UNCHANGED (intrinsic property)\n');
264 fprintf('    - Rate constant k: INCREASES\n');
265 fprintf('    - Can calculate activation energy E_a from Arrhenius plot\n');
266 fprintf('    - Irreversible reaction: extent unchanged, just faster\n\n');
267

```


Command Window

=== PROBLEM P7-7A ===

Reaction: $(\text{CH}_3)_2\text{O} \rightarrow \text{CH}_4 + \text{H}_2 + \text{CO}$
 Constant-volume batch reactor, 504°C

(a) Estimating initial pressure at $t = 0$:

Stoichiometry: $1 \text{ DME} \rightarrow 1 \text{ CH}_4 + 1 \text{ H}_2 + 1 \text{ CO}$ (net $\Delta n = +2$)

$$P(t=\infty) = 3 \times P_{\text{DME}}(t=0)$$

$$P_{\text{DME}}(t=0) = P(t=\infty)/3 = 310.3 \text{ mmHg}$$

(b) Determining reaction order and rate constant:

Time (s)	P_{total} (mmHg)	P_{DME} (mmHg)
0	310.3	310.3
390	408	261.50
777	488	221.50
1195	562	184.50
3155	799	66.00

First-order: $k = 0.000494 \text{ s}^{-1}$, $R^2 = 0.995191$

Second-order: $k = 0.000004 \text{ mmHg}^{-1} \cdot \text{s}^{-1}$, $R^2 = 0.087083$

CONCLUSION: Reaction is FIRST-ORDER

Rate constant $k = 0.000494 \text{ s}^{-1}$ ($4.9439\text{e-}04 \text{ s}^{-1}$)

Half-life $t_{1/2} = \ln(2)/k = 1402.0 \text{ s}$

(c) Suggested experimental conditions for more data:

- More points at early times (50-500 s): Better initial rate
- Intermediate times (1000-2000 s): Verify linearity
- Use higher precision pressure gauge
- Verify constant volume and temperature control
- Replicate measurements to assess uncertainty
- Check for side reactions or catalytic effects

(d) Running at higher temperature:

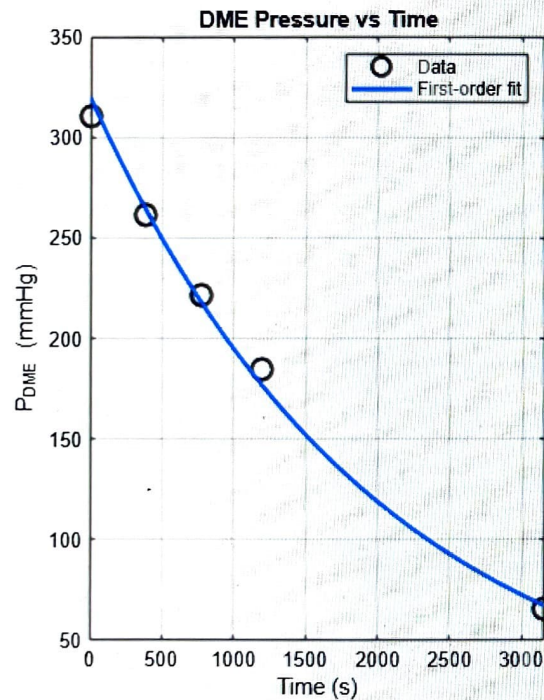
DATA CHANGES:

- Reaction rate k increases exponentially (Arrhenius)
- Same pressures reached at earlier times
- $P(t=\infty) = 931 \text{ mmHg}$ unchanged (equilibrium position)
- Would need shorter time intervals



...





P7-7A: Dimethyl Ether Decomposition

